CSE 609: Introduction to Cryptography & Network Security

Lecture – 4

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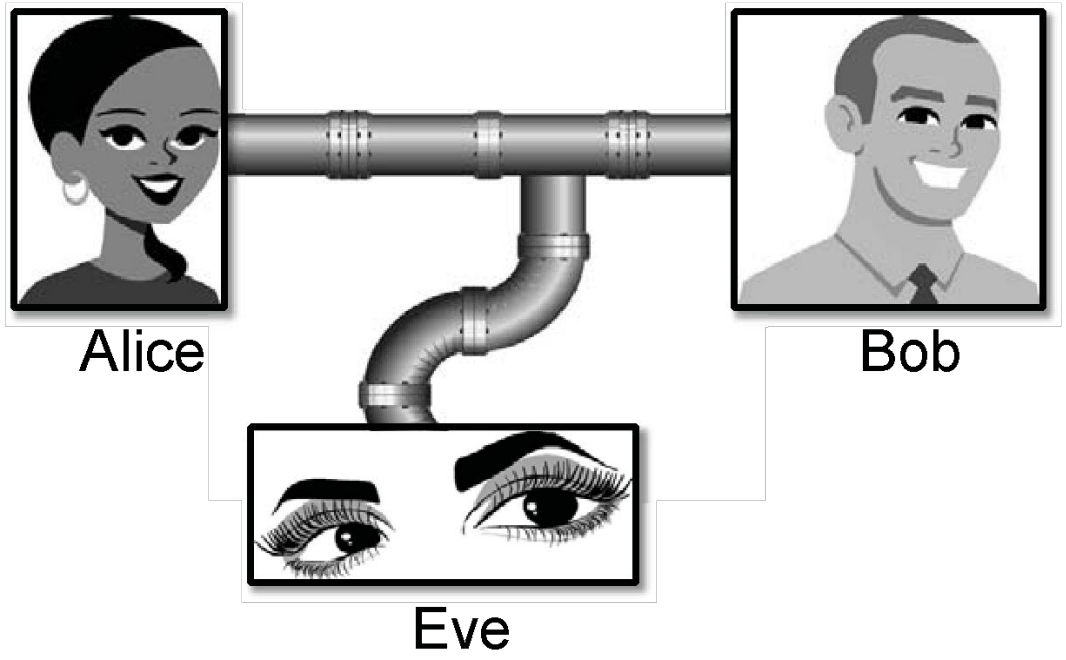
Outline

- Basic crypto concepts

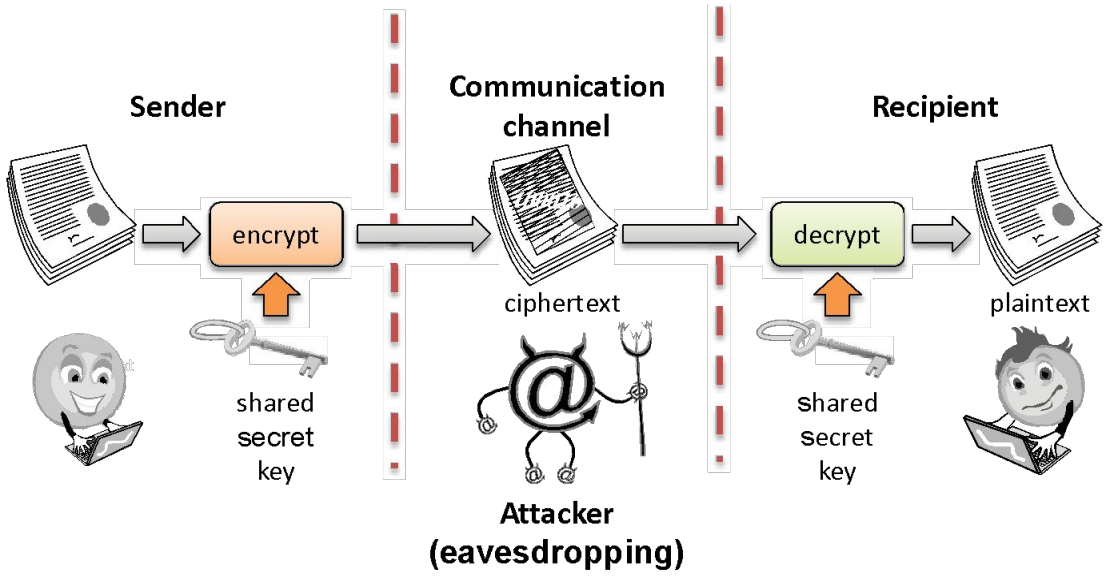
Encryption

- A means to allow two parties, customarily called Alice and Bob, to establish confidential communication over an insecure channel that is subject to eavesdropping



Encryption

- The message M is called the **plaintext**
- Alice will convert plaintext M to an encrypted form using an encryption algorithm E that outputs a **ciphertext C for M**



Encryption

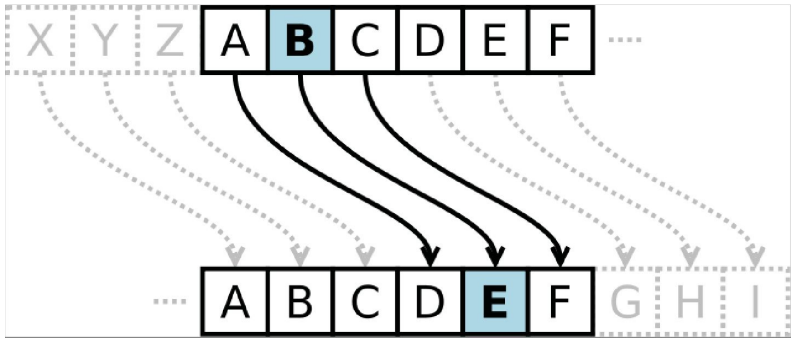
- As equations:
  - $C = E(M)$, $M = D(C)$
- The encryption and decryption algorithms are chosen so that it is infeasible for someone other than Alice and Bob to determine plaintext M from ciphertext C
- Thus, ciphertext C can be transmitted over an insecure channel that can be eavesdropped by an adversary
- The decryption algorithm must use some secret information known to Bob, and possibly also to Alice, but no other party
  - using an auxiliary input a secret number or string called ***decryption key***
  - the decryption algorithm itself can be implemented by standard, publicly available software and only the decryption key needs to remain secret
- Similarly, the encryption algorithm uses as auxiliary input an ***encryption key***, which is associated with the decryption key
- If it is feasible to derive the decryption key from the encryption key, the encryption key should be kept secret as well

Cryptosystem

- A ***cryptosystem*** consists of seven components:
  - The set of possible plaintexts
  - The set of possible ciphertexts
  - The set of encryption keys
  - The set of decryption keys
  - The correspondence between encryption keys and decryption keys
  - The encryption algorithm to use
  - The decryption algorithm to use

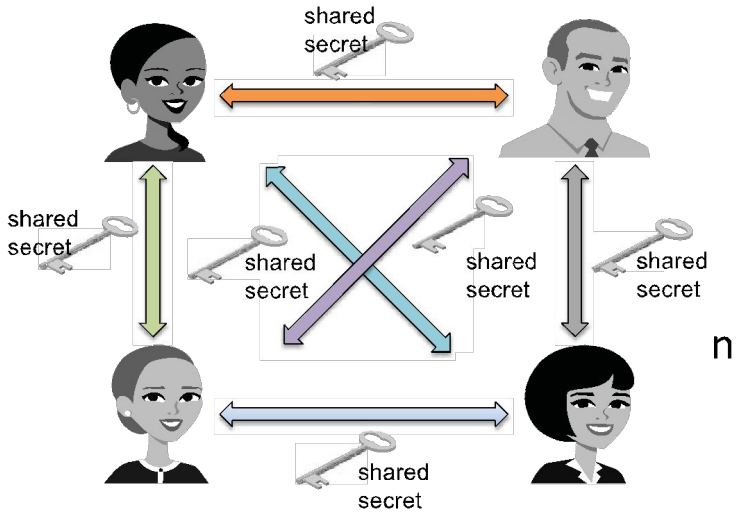
Cryptosystem: Caesar cipher

- Replace each letter with the one “three over” in the alphabet
  - An example of a shift cipher
- Can be denoted using the following formula:
  - $s(c, k)$, here c represents one of 23 letters in Latin Alphabet and k represents the key
  - If $k > 0$, forward shift (encryption) and if $k < 0$, backward shift (decryption)
  - Example: s(A,3) = D, s(D,-3) = A
- For Caesar cipher, $\{k=3\}$ is the set of encryption key and $\{k=-3\}$ is the set of decryption key
- Uses modulo operation in a sort of circular shift fashion when shift goes beyond the limit



Symmetric key distribution

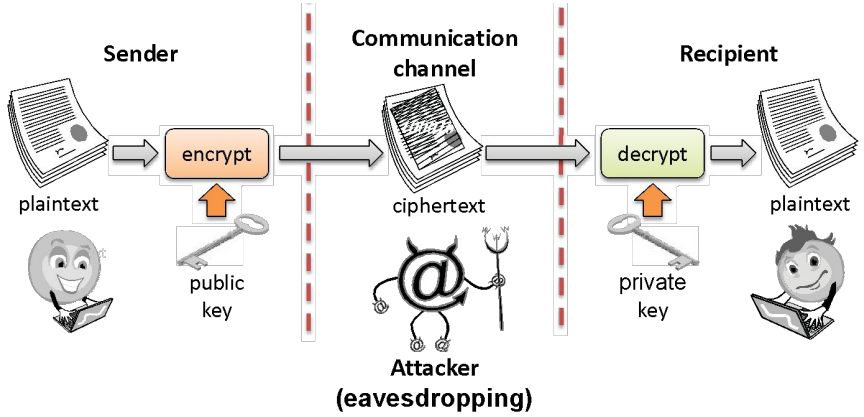
- Symmetric crypto-systems are quite fast and efficient
- However their main problem is key distribution
- It requires each pair of communicating parties to share a (separate) secret key
- If there are n parties, it means it requires a a total of $n(n-1)/2$ keys



n (n–1)/2 keys

Public key cryptography

- Bob has two keys: a **private key,** S_B, which Bob keeps secret, and a **public key,** P_B, which Bob broadcasts widely
- In order for Alice to send an encrypted message to Bob, she needs only obtain Bob's public key, P_B, and use that to encrypt her message, M, and send the result, $C = E_{P_B}(M)$, to Bob.
- Bob then uses his secret key to decrypt the message as $M = D_{S_B}(C)$
- That is, separate keys are used for encryption and decryption

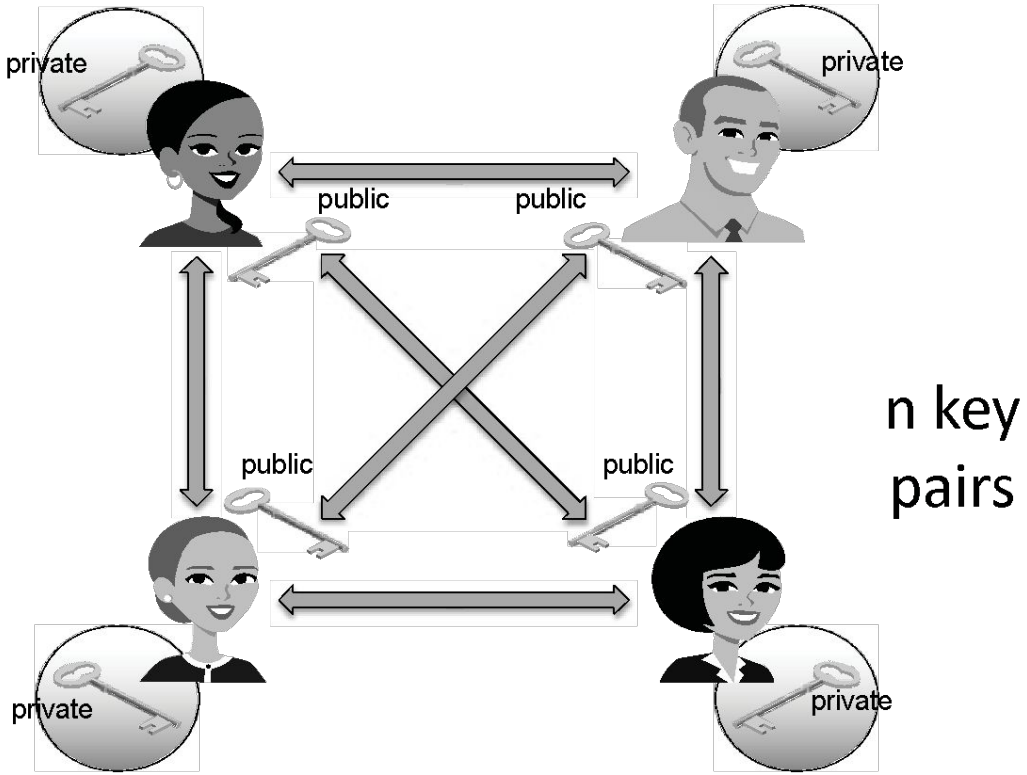


Public key distribution

- How many keys do we need to distribute for n users??

Public key distribution

- Only one key is needed for each recipient!



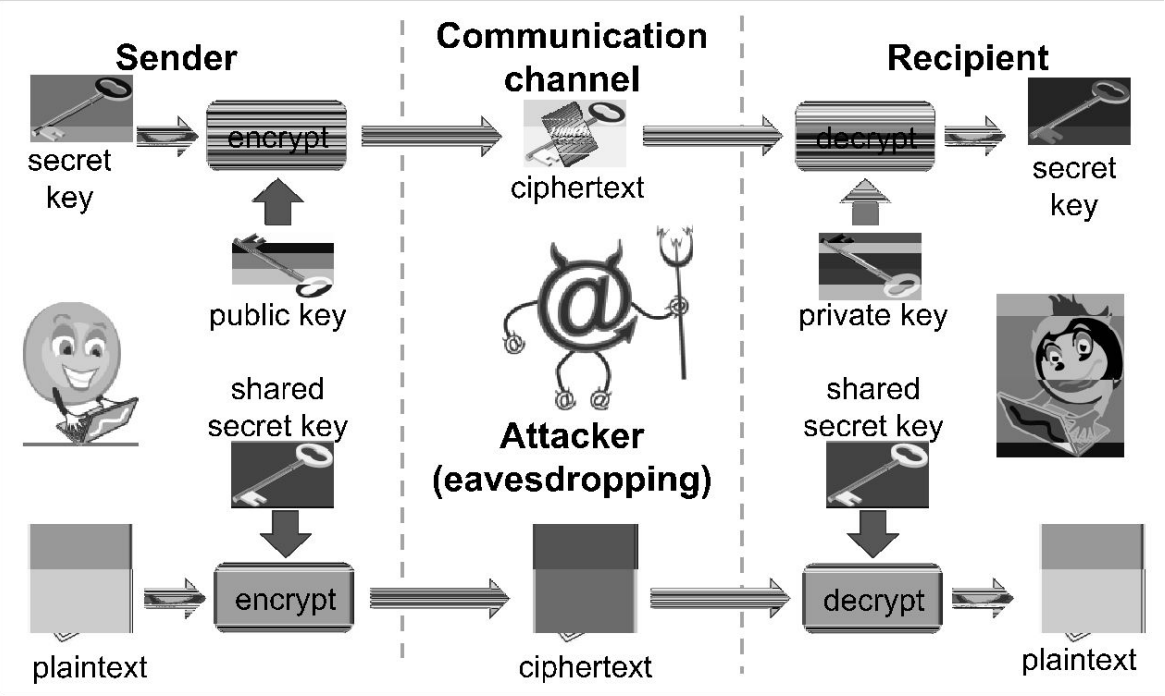
n key pairs

Combining symmetric and public key systems

- The main disadvantage of public-key cryptosystems is that they are much slower than existing symmetric encryption schemes
  - Hence, public-key cryptography is unsuitable for interactive sessions that use a lot of back-and-forth communication
- Also, public-key cryptosystems have larger key lengths than that for symmetric cryptosystems
  - For example, RSA is commonly used with 2,048-bit keys while AES is typically used with 256-bit keys

Combining symmetric and public key systems

- To work around these disadvantages,
  - public-key cryptosystems are often used in practice just to allow Alice and Bob to exchange a shared secret key
- Once exchanged, the shared secret key then can be used for communication encrypted with a symmetric encryption scheme

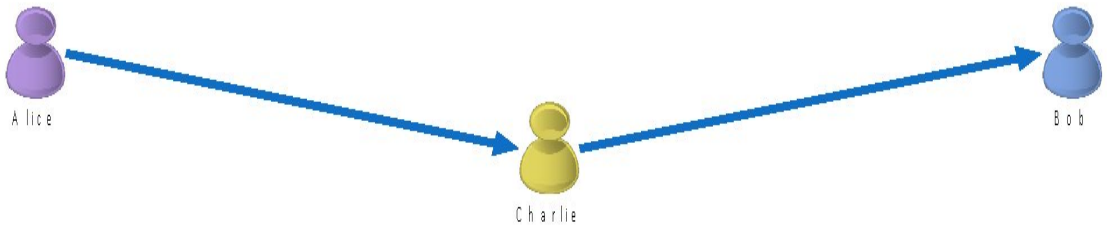


Digital signature

- Public key crypto systems allow the reversing the order in which encryption and decryption is carried out
  - $E_{P_B}\left(D_{S_B}(M)\right) = M$
- This concept is leveraged to create digital signatures
- To sign a message, M, Alice just decrypts it with her private key, S_A, creating $C = D_{S_A}(M)$
- Anyone can encrypt this message using Alice's public key, as $M' = E_{P_A}(C)$ and then compare if $M = M'$
  - If they match, the signature is valid
- Indeed, no one but Alice, who has private key S_A, could have produced such an object C, so that $M' = E_{P_A}(C)$
- Only disadvantage, the signature is as long as the message!
  - Compare it with real life signatures!

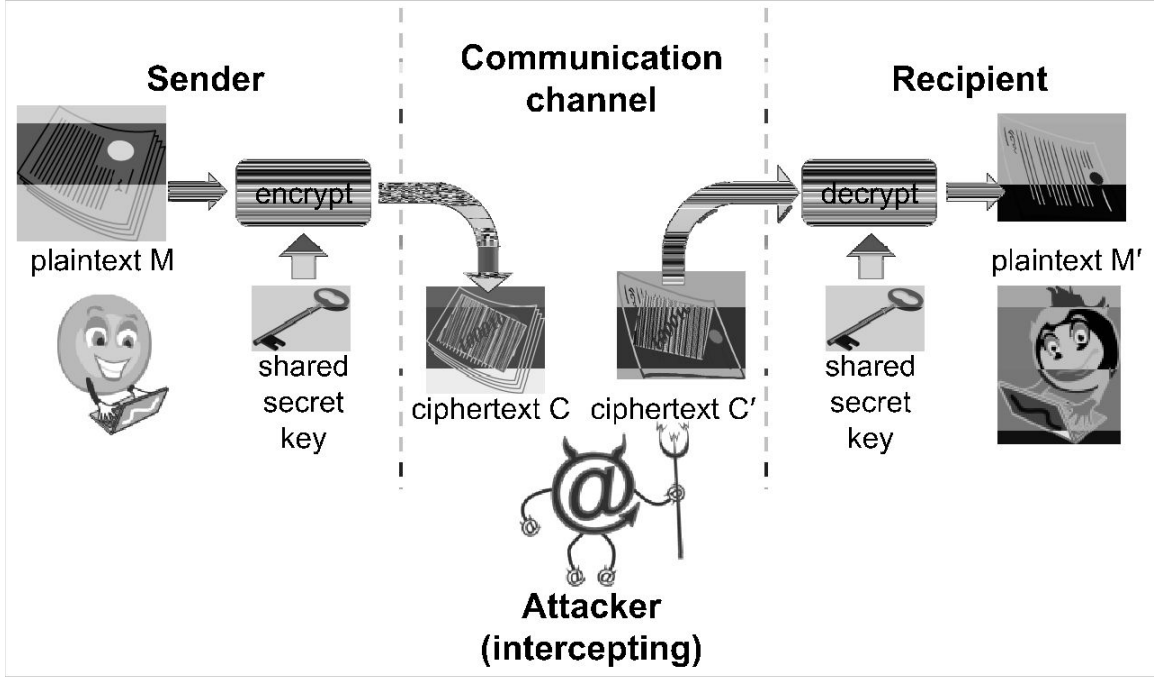
Man In The Middle (MITM) Attack

- Charlie is in the middle between Alice and Bob
- Charlie can:
  - View traffic
  - Change traffic
  - Add traffic
  - Delete traffic
- Charlie could be:
  - Internet service provider
  - Virtual Private Network (VPN) provider
  - WIFI provider such as a coffee shop
  - An attacker re-routing your connection
  - An incompetent admin (it happens)



Simple attacks on crypto systems: MITM

- The straightforward use of a cryptosystem presented, which consists of simply transmitting the ciphertext, assures confidentiality
- However, it does not guarantee the authenticity and integrity of the message if the adversary can intercept and modify the ciphertext
- How the recipient can be assured that the message s(he) receives is the intended one?



Man-In-The-Middle

Simple attacks on crypto systems: MITM

- Bob (M, S): (C,S) -> Eve (C’,S’) -> Alice (C’, S’): (M’,S’)??
- Note that *M*′ will be different from the original message *M*
  - When Alice verifies the digital signature *S*′, she obtains message *M*′ by encrypting *S’*
  - Thus, Alice is led to believe that Bob has signed *M*′ instead of *M*
- Note that in the above attacks the adversary can arbitrarily alter the transmitted ciphertext or signature
- However, the adversary cannot choose, or even figure out, what would be the resulting plaintext since she does not have the ability to decrypt
- Thus, the above attacks are effective only if any arbitrary sequence of bits is a possible message
- This scenario occurs, for example, when a randomly generated symmetric key is transmitted encrypted with a public-key cryptosystem

Simple attacks on crypto systems: brute-force

- How about brute forcing a crypto system?
  - Trying different probable keys over a cipher text to decrypt it to a meaningful text
- Mind you, encryption and decryption functions are open, just the key is secret!
- If the plaintext is an arbitrary binary string, this attack cannot succeed, as there is no way for the attacker to distinguish a valid message
- However, if the plaintext is known to be text in a natural language, then the adversary hopes that only a small subset of the decryption results (ideally just a single plaintext) will be a meaningful text for the language

Brute force attack

- Some knowledge about the possible message being sent will then help the attacker pinpoint the correct plaintext
- Key should be a sufficiently long random value to make exhaustive search attacks unfeasible
- Problem is it is usually easy to recognize that a message is a valid plaintext
  - For example, given a certain ciphertext, if an attacker could decrypt it with a given key and get message NGGNPXNGQNJABAVEIVARORNPU, which she can immediately dismiss
  - But if she gets message ATTACKATDAWNONIRVINEBEACH, then she can be confident she has found the decryption key
- This ability is related to the ***unicity distance*** for a cryptosystem

Brute force attack

- English text is typically represented with 8-bit ASCII encoding
- A message with t characters has n bits, with n = 8t
- The total number of possible *n*-bit (or t-byte) arrays is $2^{8t} = 2^n$
- It is estimated that each character of English text carries about 1.25 bits of information, known as the entropy of English
- The number of *t*-byte arrays that correspond to English text:
  - $2^{1.25t} = 2^{1.25n/8} \approx 2^{0.16n}$

Brute force attack

- In general, for some constant $0 < \alpha < 1$, there are $2^{\alpha n}$ valid text messages among the 2^n possible plaintexts
  - as not all combinations are valid messages
- The probability that a randomly selected plaintext corresponds to meaningful text is represented with: $\frac{2^{\alpha n}}{2^n} = \frac{1}{2^{(1-\alpha)n}}$
- The fraction of valid messages tends rapidly to zero as *n* grows

Brute force attack

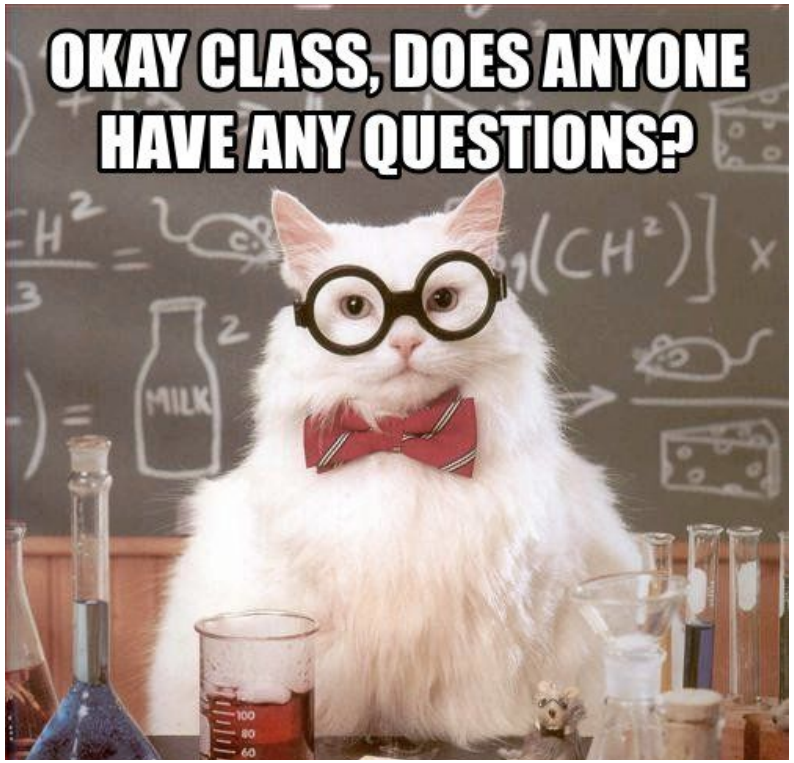
- Let k be the length (number of bits) of the decryption key
- For a given ciphertext, there are 2^k possible plaintexts, each corresponding to a key
- From the previous discussion, each such plaintext is a valid text message with probability $\frac{1}{2^{(1-\alpha)n}}$
- Hence, the expected number of plaintexts corresponding to valid text messages is $\frac{2^k}{2^{(1-\alpha)n}}$
- As the key length k is fixed, the above number tends rapidly to zero as the ciphertext length n grows

Brute force attack

- We expect that there is a unique valid plaintext for the given ciphertext when $n = \frac{k}{1-\alpha}$
- The above threshold value for *n* is called the ***unicity distance*** for the given language and key length
- The ***unicity distance*** for a cryptosystem is the minimum number of characters of ciphertext that are needed so that there is a single intelligible plaintext associated with it
- Consider a cipher text: FJKFPO that is encrypted with a substitution cipher (discussed later) and brute force might result in the following: ***thatis ofyour season onyour thatwe thetop thetwo oxford thatin thatof***
  - Which is the correct one? It is impossible to know!

Brute force attack

- For the English language and the 256-bit AES cryptosystem, the unicity distance is about 304 bits or 38 ASCII-encoded characters
  - This is only half a line of text
- We can conclude that brute-force decryption is likely to succeed for messages in natural language that are not too short
- Namely, when a key yields a plaintext that is a meaningful text, the attacker has probably recovered the original message


OKAY CLASS, DOES ANYONE
HAVE ANY QUESTIONS?
MILK